ADR-0002: Opponent Model Assumptions

ADR-0002: Opponent-model assumptions in the Tier 2 / Monte Carlo engines

Status: accepted (as a documented placeholder, not a final calibration) Date: 2026-07-23

Context

Tier 1 (`engine/fj_wager.py`) needs no assumptions about opponents beyond the legal wager rules – it computes a worst-case guarantee. Tier 2 (`engine/fj_bestresponse.py`) and the Daily Double engine (`engine/dd_wager.py`, `engine/simulate.py`) need actual probabilities: how likely is an opponent to wager a given amount, how likely are they to answer correctly, how likely is any given remaining clue to be answered correctly by someone.

ADR-0001 already rules out shipping the jwolle1 historical dataset, and this project has not yet done the deeper calibration work of mining the Tesauro et al. (2013) paper's full tables and figures (Figure 3's opponent wagering scatter plots, the six-regime empirical model for B, the round/row-by-row DD accuracy breakdown). Building Tier 2 and the DD engine now, ahead of that calibration work, means picking some numbers.

Decision

Ship illustrative, clearly-labeled placeholder parameters rather than wait:

- Opponent Final Jeopardy wagers: uniform distribution over that opponent's own Tier 1 breakpoints (bet nothing, their cover wager if one exists, their lock-tie wager if one exists, bet everything). This anchors the guesses on state-appropriate dollar amounts instead of arbitrary numbers, while being explicit that the *weights* (currently uniform) are not fitted to anything.
- Opponent Final Jeopardy correctness: 50%, flat, for every opponent and every category.
- Regular-clue success rate: 60% probability that some player answers a given remaining clue correctly, split evenly among players if so.
- Future (not-yet-found) Daily Doubles: a random player finds it and wagers 50% of their score at 60% accuracy.

Every one of these constants is named `DEFAULT_...` in code and called out in module docstrings as a placeholder, not a fitted value. Callers can override every one of them.

Consequences

- Recommendations from Tier 2 and the DD engine are directionally useful (they correctly capture *shape* – e.g., don't overbet past the point where extra money stops helping – see the "overbetting" test in `tests/test_fj_bestresponse.py`) but the exact recommended dollar amounts and win probabilities should not be quoted as validated numbers in the white paper or deck without a follow-up calibration pass.
- Any shipped claim of the form "this wager wins X% of the time" must cite which opponent-model assumptions produced it, since changing the defaults changes the number.

- Follow-up work (tracked in RISK_REGISTER.md, risk #5): mine Tesauro et al.'s published tables more thoroughly, and/or use the jwolle1 dataset internally (per ADR-0001, not for shipping) to fit real distributions, then update these defaults and re-run the test suite, which should still pass since the hand-derived tests use explicit fixed models, not the defaults.